

**Streamlining Innovation: Intellectual Property Management
System (IPMS)**

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Chapter 1

INTRODUCTION

Context of the study

Intellectual property (IP) represents a critical asset that must be effectively safeguarded to enable inventors and creators to legally and justly utilize their designs and innovations. Bently et al. (2022) state that intellectual property law encompasses various forms of creation, including inventions, literary and artistic works, as well as commercial symbols and designs. This law emphasizes that intellectual property should be protected and treasured to ensure that every inventor's designs and creations are used legally and fairly. It categorizes these creations into two primary types: Industrial Property and Copyright. Industrial Property includes inventions protected by patents, trademarks that safeguard distinctive signs, industrial designs that protect the visual appearance of products, and geographical indications that highlight products from specific regions known for their unique qualities. On the other hand, Copyright covers literary works such as novels and poems, artistic creations like drawings and sculptures, and the rights of performing artists, producers of sound recordings, and broadcasters. Each category requires distinct processes for ownership and protection, ensuring that creators' rights and benefits

are preserved. Those categorized creations can be owned through a series of processes facilitated by the law to safeguard the benefits associated with owning such creations.

The rise of the metaverse, a virtual environment for interaction and commerce—presents intricate legal issues regarding IP management. Kalyvaki (2023) points out the challenges of implementing existing intellectual property, privacy, and jurisdictional laws in this digital setting. The differing legal frameworks and data collection methods within the metaverse demand a multifaceted strategy, highlighting the need for interdisciplinary collaboration to form a thorough legal framework.

Xavdarov (2022) examines the vital function of the digital economy and the impact of IP protection on its development. Research indicates that stronger IP rights often correlate with increased innovation and economic growth; however, Neves et al. (2021) discovered discrepancies in this correlation, particularly between developed and developing countries. Their statistical evaluation suggests that while effective IP protection promotes innovation, its effects differ based on the economic environment.

Grinaldi et al. (2021) study how IP management affects a company's ability to pursue outbound open innovation.

Their findings reveal three IP strategies: defensive, collaborative, and impromptu. Interestingly, firms using a defensive strategy tend to be more involved in outbound open innovation, while those with collaborative strategies report better innovation outcomes.

As organizations experience digital transformation, the significance of workflow automation becomes increasingly clear. Ray (2023) highlights its role in improving efficiency and minimizing errors in administrative tasks. By automating routine processes, organizations can allocate human resources toward strategic initiatives, thereby boosting productivity and customer satisfaction.

The management of intellectual property within research organizations is also progressing. Kolodyazhnaya et al. (2020) support the idea of using predictive models to navigate shifts in IP management caused by digital developments. Their research suggests a fuzzy logic approach to tackle uncertainties and provide insights into effective IP management techniques.

In terms of quality management, Moisescu et al. (2024) propose a Quality Management System (QMS) designed specifically for intellectual property organizations, aiming to improve service quality and compliance with industry standards. Similarly, Novikov et al. (2024) advocate for innovative strategies in IP management

within Russia, encouraging the use of automated systems to promote commercialization in educational settings.

IBM's established leadership in patent applications highlights the strategic importance of intellectual property management. Sierotowicz (2020) analyzes IBM's experience and recommends a new IP management framework based on complexity theory to enhance economic outcomes from IP endeavors.

The combination of project management techniques with intellectual property rights is essential for successful innovation. Yeten et al. (2024) emphasize the significance of IPR in resource management and project execution, confirming that optimal practices in both fields can strengthen competitive advantages.

The increasing importance of creativity in society stresses the necessity for strong support systems for creators, as observed by Ploman et al. (2024). Asri et al. (2020) further demonstrate the significance of IP rights for small and medium enterprises (SMEs), especially in promoting economic growth through initiatives supported by the government.

Drafler (2024) discusses the interaction between IP law and antitrust law in fostering consumer welfare via innovation and competition. In the realm of decentralized finance, Bamakan et al. (2022) investigate the potential of non-fungible tokens (NFTs) in representing

intellectual property assets, suggesting frameworks to improve market access for innovators. Camarines Norte State College (CNSC) is dedicated to promoting innovation and research, resulting in an expanding collection of intellectual property assets. To fully leverage these assets, a thorough Intellectual Property Management System (IPMS) is crucial. The Intellectual Property Management Office (IPMO), created to manage CNSC's IP assets, seeks to improve its management abilities through initiatives such as the IGNITE program and partnerships with the Department of Science and Technology.

Although there are ongoing efforts, challenges still exist in effectively protecting and commercializing CNSC's innovations. This capstone project addresses these issues by creating a strong IPMS aimed at optimizing the management process of intellectual property, from inception to commercialization. Based on the research conducted by Deel (2023), this initiative aspires to bolster CNSC's capacity to safeguard its intellectual property, attract investments, and further its research goals.

The subsequent chapters will outline the design, development, implementation, and assessment of the recommended IPMS, supporting CNSC's mission of achieving excellence in research and making a significant societal impact.

Research Objective

The main objective of this research is to develop a web-based Intellectual Property Management System. Specifically, this study aimed to:

1. Determine the current system at the Intellectual Property Management Office.
2. Determine the key features and innovations of Intellectual Property Management System; and
3. determine the level of effectiveness of "Intellectual Property Management System" using the ISO/IEC 25010 questionnaires in terms of:
 - a. Functional Suitability;
 - b. Performance Efficiency;
 - c. Compatibility;
 - d. Usability;
 - e. Security;

Research Questions

This study aims to develop a Web-based Intellectual Property Management System, Specifically, this study seeks to answer the following questions:

1. What is the current system in the Intellectual Property Management Office?
2. What are the core functionalities and technological innovations that are integrated into the

Intellectual Property Management System to enhance efficiency and effectiveness? and

3. Determine the level of effectiveness using the ISO/IEC 25010 Tool in the Intellectual Property Management System in terms of:

- a. Functional Suitability;
- b. Performance Efficiency;
- c. Compatibility;
- d. Usability;
- e. Security;

Scope & Limitation

This initiative aims to create and implement a web-based Intellectual Property Management System (IPMS) for the Intellectual Property Management Office at Camarines Norte State College (CNSC). The scope of the Intellectual Property Management System (IPMS) is to manage applications for Industrial Design, Copyright, Trademark, and Utility Model, excluding full Patent applications. The system facilitates the submission of applications, simplified processes for inventors and the Intellectual Property Management Office (IPMO). It provides a centralized database for storing all intellectual property information. The system features a case management module that tracks submissions, statuses, and generates necessary documentation. Additionally, it

enhances communication between inventors and IPMO staff through integrated messaging and notification features, while offering reporting capabilities for generating reports related to submissions and approvals. However, the IPMS also has several limitations. The system does not function as an online registration platform; instead, it relies on downloadable forms with a dedicated Submission Portal for each IP category. IPMS also focuses exclusively on new submissions, which restricts its application to renewal processes or the management of existing intellectual property. The system lacks features for renewing intellectual property or online registration, narrowing its functional scope. The reliance on manual review by IPMO personnel for submissions can slow down the approval process compared to more automated systems. Reporting capabilities are limited to basic functions and do not include advanced analytics or custom report generation. Also the IPMS does not currently have a dedicated mobile application, hindering access for users who prefer mobile engagement.

Additionally, it supports only a restricted range of file formats, requiring users to convert files before uploading, which can be inconvenient. Lastly, the current user role structure is basic, with plans for more complex access controls in future updates. These factors highlight both the capabilities and areas for improvement

within the IPMS.

Significance of the Study

This study represents a major step for the institution, IPMS serving as a digital infrastructure significant because it not only simplifies administrative workflows but also acts as a strategic step for institutional innovation. By transitioning from manual records to a digital environment, CNSC ensures the long-term preservation, legal protection, and eventual commercialization of intellectual assets. The broader impact of this research is detailed through its benefits to the following stakeholders:

For the Administrators, the IPMS introduces the Administrator access maintains sovereign control over user registries and the storing of sensitive IP data. By providing monitoring tools, role-based access controls, and audit trails, the system empowers the administrator to maintain a secure and legally compliant environment. This structured oversight minimizes the risk of data loss or unauthorized access, thereby upholding the institutional reputation of CNSC as a secure hub for research and development.

For the Specialists, the system serves as a specialized workstation that enhances the precision and speed of the evaluation process. The IPMS replaces traditional, time-consuming review methods with a centralized queue

that allows specialists to systematically validate invention disclosures across various IP categories. The significance of the case management module cannot be overstated; it provides a real-time communication bridge, allowing specialists to issue immediate feedback and track the progress of revisions. This efficiency not only optimizes the specialist's workload but also ensures that high-potential ideas are processed quickly, reducing the "time-to-protection" and facilitating a more agile transition from laboratory or classroom concepts to legally recognized intellectual property.

For the Inventors, including faculty, staff, and students, the Intellectual Property Management System represents a democratization of access to intellectual property rights through a user-centric interface that lowers barriers to entry for protecting creative and technical works. Inventors gain autonomy to monitor their application's journey, access standardized Intellectual Property reference materials, and engage in direct, transparent communication with reviewers. This engagement is crucial for fostering a sustainable culture of innovation within the organization. When the path to Intellectual Property protection is clear and accessible, it motivates researchers to pursue more ambitious projects.

Chapter 2

REVIEW OF RELATED LITERATURE

This literature review explores three main objectives essential for developing a robust Intellectual Property Management System (IPMS) for Camarines Norte State College (CNSC). The objectives include assessing the current system at the Intellectual Property Management Office (IPMO), analyzing key features and innovations in existing IPMSs, and evaluating effectiveness using the ISO/IEC 25010 framework. Evaluation criteria will focus on functional suitability, performance efficiency, compatibility, usability and security. Insights from this review will be crucial in creating a reliable and user-friendly system that enhances innovation and maximizes the value of CNSC's intellectual property.

The management of intellectual property (IP) has become increasingly vital for organizations that aim to innovate and maintain a competitive advantage in today's dynamic environment. This is particularly relevant for the IPMO, which oversees the creation, registration, and enforcement of IP rights. Currently, the IPMO employs a traditional paper-based system for its operations, emphasizing compliance through various documentation forms. While this system offers some benefits, such as legal recognition and staff familiarity, it also presents significant challenges that impede efficiency and overall

effectiveness. A major drawback of the paper-based system is the cumbersome nature of physical document storage, which can be both space-consuming and inefficient. The manual handling of documents often results in delays when accessing critical information, as staff may need to conduct time-consuming searches through filing cabinets. Although the office utilizes some basic digital tools, such as email and file management systems, these tools are limited in functionality and do not provide the comprehensive capabilities needed for effective IP management. Therefore, it is essential to evaluate the current system's effectiveness and its limitations compared to a more modern web-based alternative.

Research by Smith et al. (2021) indicates that a web-based system can provide instant access to documents from any location, significantly improving efficiency and fostering collaboration among staff. Furthermore, while basic digital tools may aid communication, collaboration on paper documents remains cumbersome, often requiring physical meetings and the distribution of multiple copies for review, which can lead to miscommunication and delays. Jones (2022) argues that a web-based system would facilitate real-time collaboration, enabling multiple users to access and edit documents simultaneously, thereby streamlining work flows and decision-making processes.

Although the current paper-based system may appear cost-effective initially, ongoing expenses related to paper supplies, printing, and physical storage can accumulate over time, ultimately making it less economical. Brown (2023) notes that a web-based system can help reduce these costs by minimizing the need for physical materials, thus enabling more efficient resource management. Additionally, while paper documents are often viewed as secure and reliable, they are vulnerable to damage, loss, or unauthorized access. Green and White (2024) assert that a web-based system enhances security through digital backups and access controls, ensuring sensitive information is well-protected and easily retrievable.

Examining the key features and innovations of an Integrated Project Management System (IPMS) reveals a comprehensive approach to enhancing organizational efficiency. Centralized databases are crucial in IPMS, facilitating data management and integrity. Kunjir and Suryawanshi (2021) emphasize the effectiveness of centralized database management systems in academic institutions, highlighting their ability to reduce redundancy and improve data accuracy. Saha and Saha (2022) further stress the importance of ensuring data integrity for informed decision-making within the IPMS framework.

Work flow management is another critical component that significantly impacts organizational performance. Zhang and Zhao (2023) provide evidence that effective work flow management can optimize processes, improve resource allocation, and enhance overall productivity. Additionally, Lee and Kim (2021) explore how work flow automation can streamline business processes by reducing manual tasks and minimizing human error. Document management is vital for the efficient organization and retrieval of information critical to managing IP assets. Moore and Johnson (2022) examine current trends in document management systems, while Patel and Singh (2023) argue that these systems can significantly enhance organizational efficiency, especially in small and medium enterprises.

Collaboration tools are essential for fostering teamwork within an IPMS. Garcia and Martinez (2021) illustrate how digital tools can enhance collaboration among team members, while Kumar and Joshi (2023) conduct a systematic review of collaborative tools specifically designed for remote work environments. The inclusion of such tools can improve communication and coordination among staff, particularly in today's increasingly hybrid work settings.

Reporting and analytics capabilities are another significant innovation in IPMS. Thompson and Patel (2022) discuss how leveraging analytics can enhance decision-making processes in intellectual property management, providing a framework for effective data utilization. Anderson and Green (2023) emphasize the importance of data reporting in improving overall organizational performance, highlighting the need for systems that deliver actionable insights.

The introduction of inventor portals can further enhance user engagement and facilitate innovation within the IPMS. Chen and Hu (2024) advocate for a user-centered design approach in developing effective inventor portals, while Roberts and Lewis (2023) discuss their positive impact on university technology transfer offices, demonstrating how such portals can promote a culture of innovation and collaboration.

User authentication is critical for ensuring security within the IPMS. Davis and Miller (2022) review various user authentication methods, emphasizing their importance in safeguarding sensitive information from unauthorized access. Additionally, real-time data processing capabilities are increasingly vital, allowing organizations to access and analyze data swiftly, thereby improving responsiveness to emerging challenges.

To effectively evaluate the performance of an IPMS, it is essential to explore the eight quality characteristics outlined in the ISO/IEC 25010 framework. This framework provides a structured evaluation approach and emphasizes the interconnections of these attributes in determining the overall effectiveness of the system. Functional suitability focuses on how well the software meets user requirements. Khan and Khan (2020) highlight the importance of systematic evaluation in ensuring user satisfaction and operational efficiency. Performance efficiency pertains to the effective utilization of resources while delivering required functionality. Jiang and Zhang (2021) stress that optimizing resource use is critical for achieving better performance outcomes. Compatibility reflects the system's ability to operate seamlessly across various environments, as discussed by García and López (2021).

Usability measures how effectively users can interact with the system. Zhang and Li (2021) emphasize user-centered design principles that enhance user satisfaction and productivity. Reliability indicates the system's ability to perform consistently over time, with Kuo and Zhang (2021) presenting a framework for assessing reliability through systematic evaluation methods. Security is paramount for protecting sensitive information, with Safi and Alzahrani (2022) advocating

for comprehensive security evaluations to identify vulnerabilities. Maintainability should be integrated into the development life-cycle, with Ali and Khan (2021) emphasizing that systems should be designed for easy updates and modifications. Lastly, portability refers to the system's ability to operate in various environments with minimal modification, as highlighted by Kumar and Singh (2021).

Synthesis of the Art

In conclusion, integrating a centralized database, effective work flow management, robust document management, and advanced collaboration tools is essential for creating an efficient IPMS. Reporting and analytics capabilities empower users to make informed decisions, while real-time data processing enhances organizational agility. By addressing these critical areas, the development of an effective IPMS for Camarines Norte State College will not only meet user needs but also significantly improve overall organizational effectiveness and efficiency.

In reviewing current research on Intellectual Property Management Systems, several shortcomings arise that our proposed system for CNSC aims to address. One notable gap in the literature is the lack of practical strategies for transitioning from conventional systems to

modern web-based alternatives. While numerous studies highlight the drawbacks of paper-based methods, they often fail to provide tangible solutions for establishing centralized databases that enhance data integrity and accessibility (Kunjir & Suryawanshi, 2021). Our IPMS will directly address this issue by implementing a strong centralized database.

Additionally, while the automation of work flows is frequently discussed, existing literature often lacks specific examples demonstrating successful implementation. Our system will incorporate automated work flows designed to simplify IP management and significantly reduce manual errors, thereby increasing operational efficiency (Zhang & Zhao, 2023).

Current discussions on document management innovations tend to focus on theoretical improvements without examining their practical application in academic settings. Our proposed IPMS seeks to bridge this gap by introducing advanced document management tools that enhance the organization and retrieval of IP-related documents (Moore & Johnson, 2022).

Collaboration is another area where existing studies recognize its significance, but often do not provide practical tools for fostering teamwork in hybrid environments. Our IPMS will incorporate extensive collaboration features that facilitate real-time

communication and project management, promoting a more cooperative work culture (Garcia & Martinez, 2021).

Furthermore, while many studies endorse the use of analytics in decision-making, they frequently fail to elaborate on detailed frameworks for execution. Our system will encompass comprehensive reporting and analytics functionalities, enabling data-driven insights that enhance strategic planning and decision-making (Thompson & Patel, 2022).

Security is a critical concern referenced in the literature, yet specific evaluations of security measures are often inadequate. Our IPMS will prioritize user authentication and data protection, ensuring that sensitive information is safeguarded against potential threats (Davis & Miller, 2022).

User engagement is vital for the efficacy of any IP management system; however, the literature discusses user-centered designs without offering actionable strategies to foster engagement. Our proposed IPMS will include an inventor portal aimed at enhancing user interaction and stimulating innovation, thus addressing this need (Chen & Hu, 2024).

Access to real-time data is recognized as crucial for agility in IP management, but many studies do not explore effective implementation methods. Our system will enable real-time data processing, improving CNSC's

capacity to respond promptly to emerging challenges (Anderson & Green, 2023). Compliance management is another critical area that existing studies occasionally address but often lack comprehensive tools for effective tracking. Our IPMS will provide features specifically designed for managing compliance with IP regulations, ensuring that CNSC meets necessary legal requirements (Ali & Khan, 2021). The need for integration with existing systems is frequently mentioned, yet practical solutions for achieving compatibility are often missing in the literature. Our IPMS will ensure seamless integration with other software utilized within CNSC, enhancing overall functionality and user experience (García & López, 2021).

Usability and user experience are crucial for the success of any system; however, while these factors are discussed, the literature often does not focus on the design principles necessary for an intuitive interface. Our proposed IPMS will emphasize user-centered design, making the interface accessible and user-friendly (Zhang & Li, 2021).

Finally, while the importance of technical support and maintainability is widely recognized in the development of information systems, many studies fall short in offering comprehensive frameworks for ongoing maintenance and support. Our Intellectual Property Management System

(IPMS) will incorporate robust support mechanisms designed to ensure both longevity and effectiveness, thereby addressing this crucial need (Kumar & Singh, 2021). These mechanisms will include regular software updates to keep the system current, user training sessions to enhance skills and knowledge, and a dedicated helpdesk staffed by knowledgeable personnel to assist users with any challenges they may encounter.

By proactively addressing these gaps, our proposed IPMS for Camarines Norte State College (CNSC) aims to create a more effective, user-friendly, and innovative system tailored to the specific needs of its users. This system will not only meet the diverse needs of faculty, students, and administrators but also significantly enhance the overall management of intellectual property through centralized data management, automated workflow processes, and comprehensive tracking capabilities. By fostering a supportive and responsive environment, we can encourage greater engagement from all stakeholders involved, reducing the anxiety and uncertainty that often accompanies the IP application process. The streamlined workflows and transparent tracking will establish best practices in institutional Intellectual Property management, positioning CNSC as a model institution that values and protects innovation.

METHODOLOGY

Research Methodology

This methodology section outlines the approach taken to develop the Intellectual Property Management System (IPMS) for Camarines Norte State College (CNSC). It describes the research design, data collection methods, and analysis techniques used to ensure the system effectively meets the college's needs.

The process includes several key phases: needs assessment, design and development, implementation planning, and evaluation. Each phase builds on the previous one to create a user-centered and effective system.

By detailing this framework, we aim to clarify how the IPMS will be developed and implemented, ultimately improving the management of intellectual property at CNSC.

Software Development Methodology

We utilize the Scrum Agile methodology for this project due to its emphasis on collaboration, flexibility, and iterative progress. The structured framework of Scrum allows for regular assessment and adaptation, which is essential for the development of the Intellectual Property Management System (IPMS). This

iterative nature enables the team to respond effectively to changing requirements and priorities, ensuring a smooth and productive development process while fostering continuous improvement.

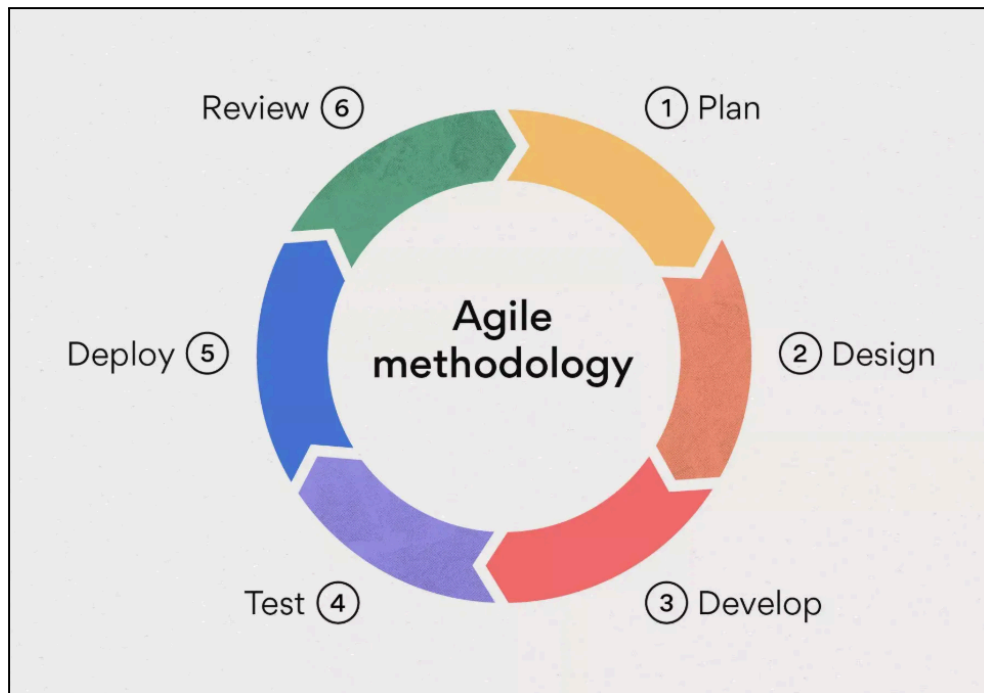


Figure 1: Agile Scrum Methodology

Schedule of Activities

This section outlines the schedule for our capstone project, divided into two parts: Capstone 1 and Capstone 2. Capstone 1 includes Chapters 1-3, which are already completed, and is detailed in Figure 2. Capstone 2 focuses on completing Chapters 4 and 5 (Results and Discussion, and Conclusion and Recommendations) and the important system development phase, as shown in Figure 3. This schedule helps us track our progress in developing the Intellectual Property Management System (IPMS) and

ensures we meet our deadlines. By breaking down tasks and setting deadlines, we can manage the project effectively, leading to a functional and user-friendly IPMS.

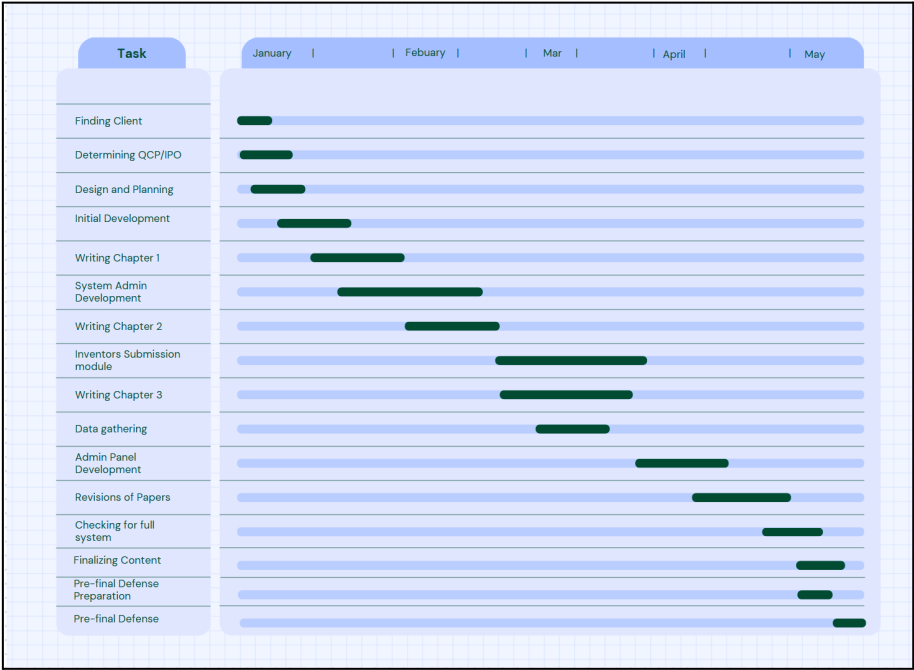


Figure 2: Schedule of Activities Capstone 1

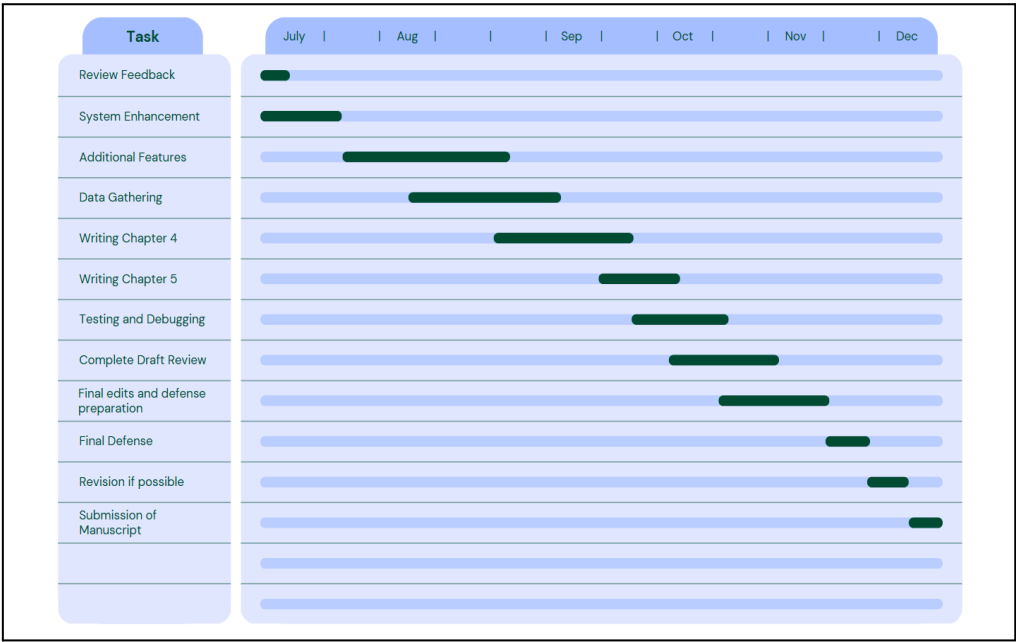


Figure 3: Schedule of Activities Capstone 2

Sources of Data

The primary information source for this research is the Intellectual Property Management Office (IPMO) at Camarines Norte State College (CNSC), situated in the COENG building. Central to this data collection was the involvement of Engr. Vega, the IPMO representative, whose specialized insights into the office's workflows and administrative procedures were indispensable. Her expertise allowed for a critical assessment of the existing system's strengths and vulnerabilities, serving as a foundational blueprint for the design and functional requirements of the proposed Intellectual Property Management System (IPMS). By identifying essential features and transaction modules through this direct collaboration, the research ensures that the resulting IPMS is not only technically sound but also practically aligned with the institutional demands of the IPMO.

Complementing these primary insights, secondary data was acquired through the administration of questionnaires to a randomly selected group of potential stakeholders, encompassing individuals both within and outside the college campus who may engage in intellectual property submission. The inclusion of this diverse demographic is intended to yield unbiased results that are generalizable to a broader population, moving beyond the limitations of the existing user base. This approach facilitates a

deeper understanding of the user experience regarding project submissions, tracking methodologies, and documentation challenges. Furthermore, the study integrates an extensive review of relevant literature, scholarly articles, and existing studies. This synthesis of secondary sources provides a benchmark for best practices and global innovations, ensuring that the development of the IPMS is informed by proven strategies and modern technological advancements.

Population of the study

The study engages 20 to 25 participants using random sampling to ensure findings represent a broader socio-technical landscape. This selection process is critical as it reduces sampling bias inherent in self-selected user groups. By randomly selecting from the general population, the data reflects diverse perspectives and experiences, providing every potential stakeholder equal probability of participation. This approach enhances research credibility and establishes a statistically sound foundation for broader inferences. The strategy invites fresh perspectives from individuals not habituated to the current system, enabling identification of latent usability issues and discovery of opportunities for system optimization.

Data Collection Method

The research employs a mixed-method approach to

ensure systematic collection and analysis of data regarding the proposed Intellectual Property Management System (IPMS). This methodology utilizes both qualitative and quantitative instruments as primary tools for data acquisition, creating a comprehensive framework that captures the multifaceted nature of intellectual property management needs. Semi-structured interviews provide in-depth insights into current operational challenges, staff experiences, and workflow inefficiencies within the IP Management Office. These qualitative conversations allow participants to articulate nuanced perspectives and identify pain points that may not be captured through numerical data alone. Concurrently, structured survey instruments and questionnaires with closed-ended questions and Likert-scale ratings allow for precise measurement of respondent satisfaction, frequency of intellectual property submissions, and perceived effectiveness of existing procedures.

The quantitative component enables statistical analysis and provides measurable benchmarks for system performance evaluation. By integrating qualitative narratives with quantifiable metrics, the study performs comprehensive evaluations to identify significant patterns and trends among the target population. This triangulation of data sources enhances the validity and reliability of findings, offering multiple perspectives

on the same phenomenon. The combined approach provides a rigorous foundation for the system's requirements and ensures that the final design of the IPMS is grounded in both empirical evidence and authentic user experiences, ultimately resulting in a solution that addresses real-world needs with precision and relevance.

Data Analysis

To ensure comprehensive evaluation of the gathered data, this study employs a mixed-method approach combining qualitative and quantitative analyses. Descriptive statistics provide systematic summaries of the raw data, focusing on measures of central tendency (mean, median, mode) and measures of variability (standard deviation, variance). These metrics characterize the distribution of respondent perceptions and establish baseline requirements for the Intellectual Property Management System (IPMS).

Semi-structured interviews with the IP Director, Technical Assistant, and support staff revealed significant operational challenges within the current manual system. Participants described the existing infrastructure as fragmented, relying on physical paperwork, disconnected email communications, and isolated digital folders. The IP Director noted that verifying application status requires extensive cross-referencing of physical files and disparate

records, creating substantial inefficiencies. This decentralized approach not only delays operations but generates uncertainty among inventors regarding the progress of their work. Staff members described operating in a predominantly reactive mode, constantly responding to status inquiries due to the absence of self-service tracking capabilities for inventors.

The second phase of analysis examined how the proposed IPMS directly addresses these operational frustrations. Rather than simply offering a technological upgrade, the IPMS fundamentally transforms workflow dynamics. Core functionalities—specifically the unified dashboard and real-time status tracking—eliminate the information opacity that currently impedes productivity. Automated activity logs create transparent audit trails, capturing every specialist review and inventor update within an accessible timeline. This transition from manual to digital workflows enables staff to shift focus from administrative searching to high-value activities: protecting and advancing institutional innovation. The data suggests the IPMS represents more than efficiency enhancement; it establishes a professional, trustworthy framework between the institution and its innovators, ensuring no valuable intellectual contribution is lost to systemic inefficiency.

Research Question 3 utilized a quantitative approach to

analyze data gathered during system development and evaluation. Descriptive statistics, centered on weighted mean calculations, determined average responses for each quality characteristic defined in the ISO/IEC 25010 framework. By calculating weighted means, raw data was converted into simplified values quantifying respondents' experiences of the system's functional suitability, performance efficiency, and security. These metrics, paired with their corresponding verbal interpretations, are essential for characterizing perceptions and validating system requirements. By synthesizing these numerical values, the researchers effectively illustrated general consensus among participants regarding the system's performance and functionality.

Following the descriptive phase, inferential statistical analyses extrapolated findings from the 20-participant sample to the broader institutional community. This phase involved hypothesis testing and confidence interval calculations to determine the statistical significance of results. By assessing the probability that observed patterns were not due to chance, the study made authoritative generalizations about user needs and system acceptance. This quantitative framework ensures that conclusions are mathematically sound and provides a robust foundation for final IPMS design and implementation. This data-driven methodology eliminates

subjective bias, ensuring system development is guided by objective performance metrics and verified user requirements.

Ethical Consideration

This study is anchored in a rigorous commitment to ethical research principles to ensure the integrity of findings and the protection of all participants. Prior to data collection, participants were provided with comprehensive information regarding the study's objectives, current operational procedures at IPMO, and the specific research purpose. To ensure informed evaluation, a guided demonstration of the proposed IPMS was conducted, including system navigation and core features explanation, culminating in written informed consent. Participation remained entirely voluntary, granting individuals the right to decline specific questions or withdraw at any stage without penalty. To uphold privacy standards, all collected data was systematically anonymized to remove personal identifiers, ensuring responses remain untraceable. The research approach is defined by honesty and transparency; findings are reported accurately, and limitations are openly acknowledged. By fostering a respectful environment that values participant perspectives, this study produces credible insights while prioritizing the welfare and dignity of every individual involved.

Chapter 4

RESULTS AND DISCUSSIONS

This chapter presents the researcher's findings from the study titled "Streamlining Innovation : Intellectual Property Management System", along with the survey, analysis and interpretation of the collected data through various evaluation and user testing. As it aims to answer the following questions: (1) "What is the current system in the Intellectual Property Management Office?" (2) "What are the core functionalities and technological innovations that are integrated into the Intellectual Property Management System to enhance efficiency and effectiveness?" and (3) "What is the level of effectiveness using the ISO/IEC 25010 Tool in the Intellectual Property Management System in terms of functional suitability, performance efficiency, compatibility, usability and security?". This chapter addresses those research questions and provides important insights into the conceptualization, design, development and evaluation of the Intellectual Property Management System for Intellectual Property Management Office (IPMO) located at Daet, Camarines Norte. By systematically addressing each research objective, this chapter illustrates the transition from the limitations of the existing manual process to the enhanced capabilities of

the developed digital solution, Intellectual Property Management System.

IPMO Intellectual Property Document Processes

To answer the first research question, "What is the current system of the Intellectual Property Management Office?" The researchers conducted a series of interviews with the staff of the Intellectual Property Management Office located in Daet to determine the existing system for processing Intellectual Property documents, management procedures, and record storage practices.

Application of Intellectual Property Rights and Claims. These applications are divided into four Intellectual Property categories: Copyright, Trademark, Utility Model, and Industrial Design. New inventors are required to submit an application form for their project submission containing their personal inventions or innovations. These forms include personal information and supporting documents necessary for evaluation.

Application Process. The application process begins when an inventor submits their inventions or innovations along with the required files and application forms. An office staff member or the Office Admin identifies a suitable Specialist or specialist to review the submissions. Inventors whose submissions are rejected for various reasons and wish to resubmit their works are required to revise all submitted documents, addressing

all remarks given by the Specialist along with the needed documentation. After reviewing and confirming no discrepancies exist, the submissions are deemed approved for filing IP rights and claims.

Manual Review Process. The staff of the Intellectual Property Management Office relies on manual review for each submission. The review process is limited to printed or hard copy files, and digital submissions such as digital 3D visualizations, video, sound, games, or digital art designs are not yet supported by the services provided by IPMO.

Reporting Process. Office efficiency reports are limited to basic documents encoded by office personnel, and no advanced data-driven analytic tools have been utilized for generating reports. Reporting capabilities are limited to basic functions without systematic tracking or trend analysis.

The findings revealed that the Intellectual Property Management Office (IPMO) uses a paper-based and manual system to process applications and project submissions, as well as for record management. Inventor and innovator information, application details, and project submission histories are recorded in file books and paper forms, which are then stored in file cabinets.

The results align with previous studies examining manual IP management systems. Sharma and Patel (2021) reported

that offices relying on manual paper-based systems experienced significant data inconsistencies and service delays, with processing times increasing by up to 40% compared to digital alternatives. Chen et al. (2022) found that incomplete documentation and the absence of structured referral processes hindered effective file management in institutional IP offices, resulting in lost applications and duplicated efforts. Martinez and Rodriguez (2020) stated that physical document transfers caused communication breakdowns between inventors and reviewers, with 35% of applicants reporting confusion about their application status. Furthermore, Thompson and Lee (2023) demonstrated that institutions utilizing manual IP management systems faced challenges in scalability, noting that as submission volumes increased, error rates rose proportionally due to human fatigue and organizational complexity. Kim and Park (2021) emphasized that the lack of centralized digital repositories in traditional IP offices created vulnerabilities in data security and made compliance auditing significantly more difficult, exposing institutions to potential legal and administrative risks.

Features of Intellectual Property Management System

To answer the second research question, "What are the core functionalities and technological innovations that are integrated into the Intellectual Property

Management System to enhance efficiency and effectiveness?" the features of the Intellectual Property Management System includes the following:

Centralized Login Interface for User and Security Control. This feature ensures that there is a user type validation system to limit access functions irrespective to their User type such as Admins, specialists, and Inventors Centralized Login Interface. This feature enables the Users to login and access their dashboards according to their level access as shown in Figure 4.

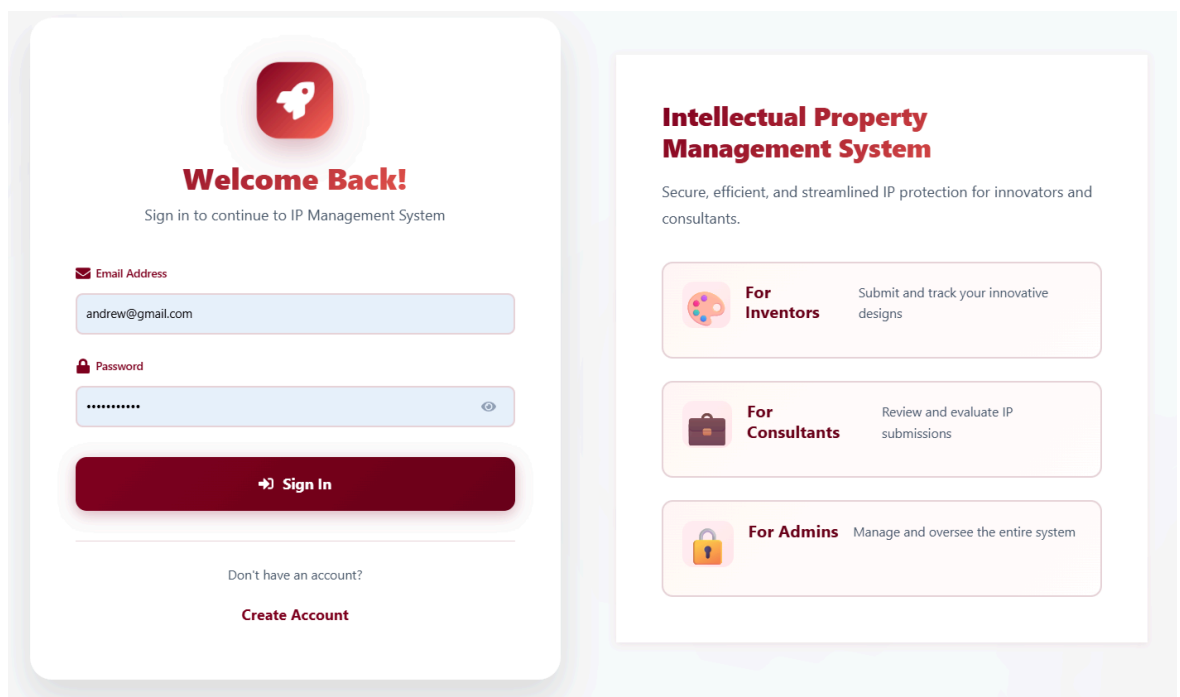


Figure 4. Login Interface

Create an Account. This feature allows New User to make their own personal Accounts with unique identification numbers and to record their initial basic information such as their name, email address and their unique Password. Users were categorized into 3 distinct

User Level Access, Admin, specialist and Inventors. This comes in a simple Infographic UI to help new Users to better understand their user types as shown in Figure 5.

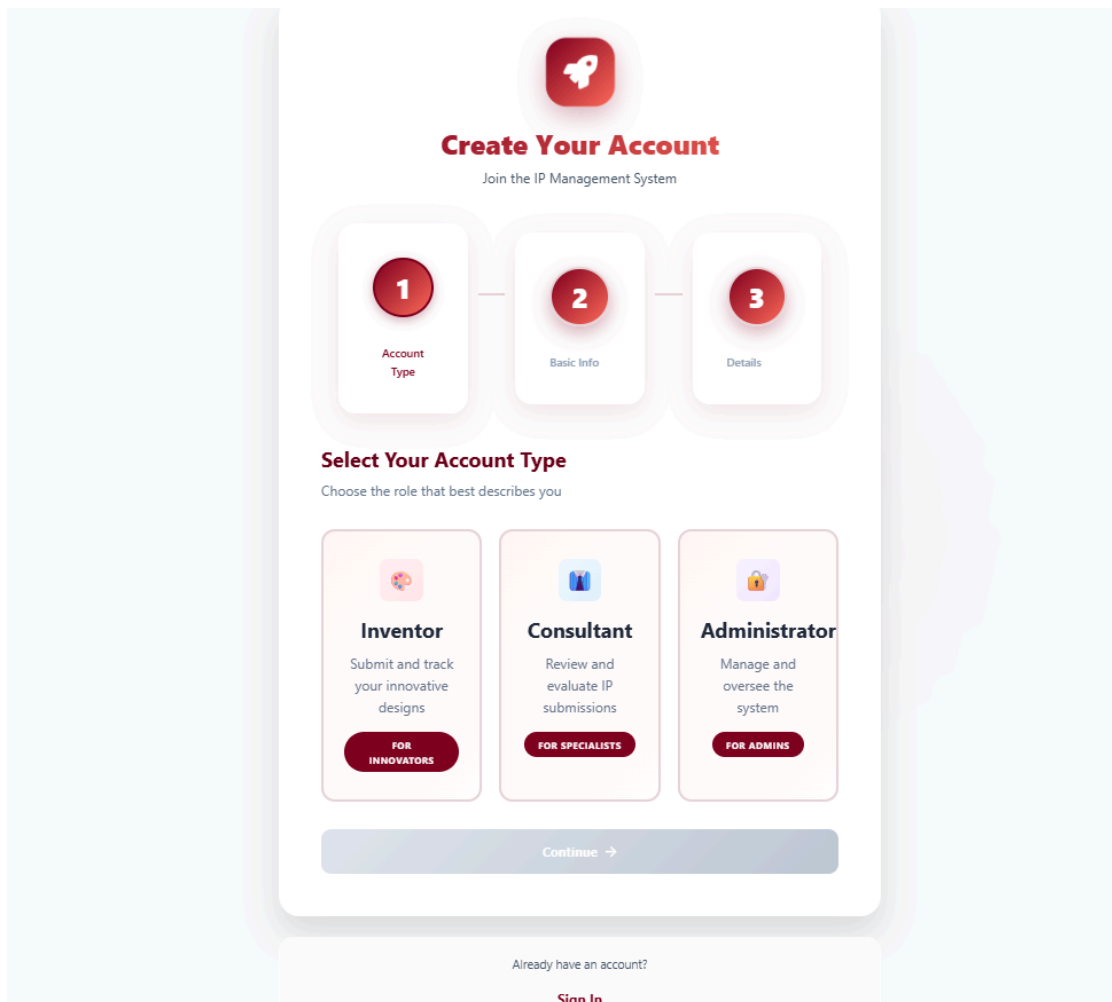


Figure 5. Sign up Interface

Admin Dashboard. This feature provides an immediate overview of system health and performance metrics, including active reviews, approval rates, and total submissions. It has detailed breakdowns of IP categories (e.g., Copyright, Industrial Design and Trademark) and a "System Summary" section to track average processing

time. The dashboard includes a "Recent Activity" feed, enabling administrators to monitor the latest transactions and ensure the system status remains operational as shown in Figure 6.

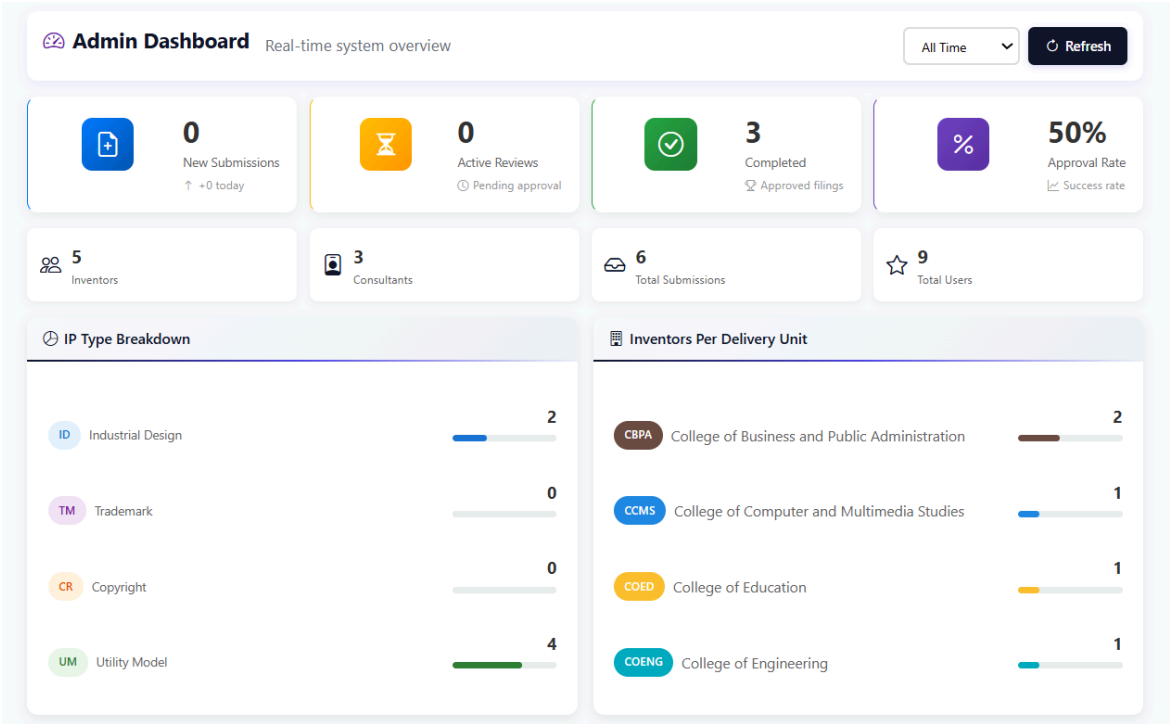


Figure 6. Admin Dashboard

User Management. A master directory for Admin Users to approve new accounts, manage existing profiles, and grant access to navigate their respective dashboards for both Inventors and Specialists. This centralized control mechanism ensures that every user is authenticated, verified as real and unique, and secured to avoid account redundancies and unauthorized access. By implementing a structured approval workflow, the system maintains data integrity and establishes accountability across all user interactions, as shown in Figure 7.

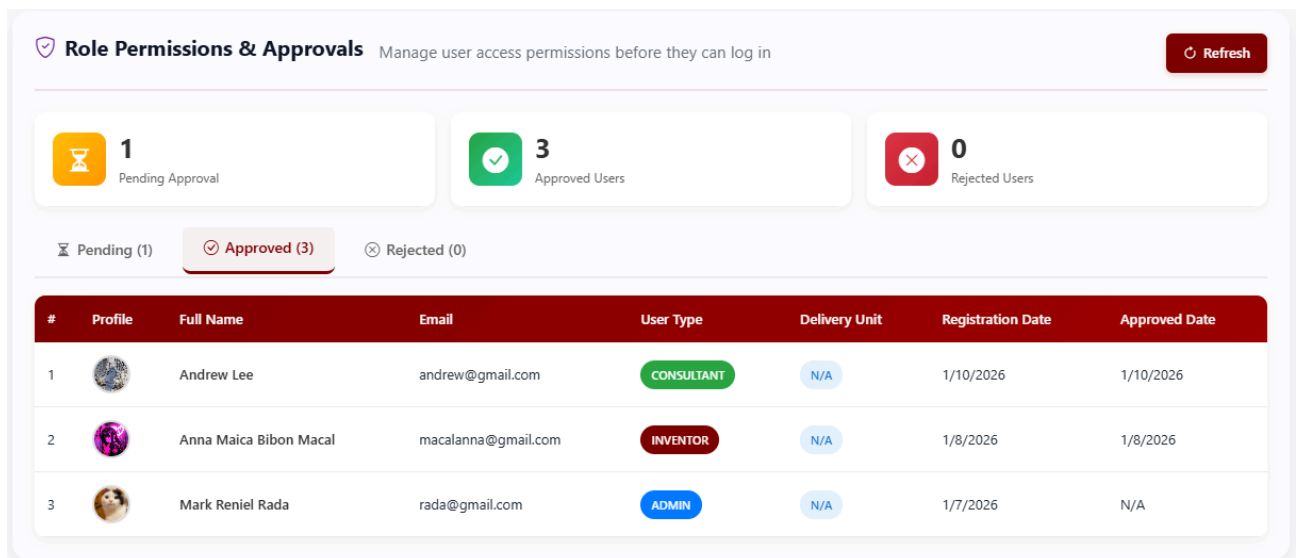


Figure 7. User Management

Project Submission Monitoring. This feature serves as an over-all system activity to ensure no submission is ignored and that Specialists are completing tasks in a timely manner as shown in Figure 8.

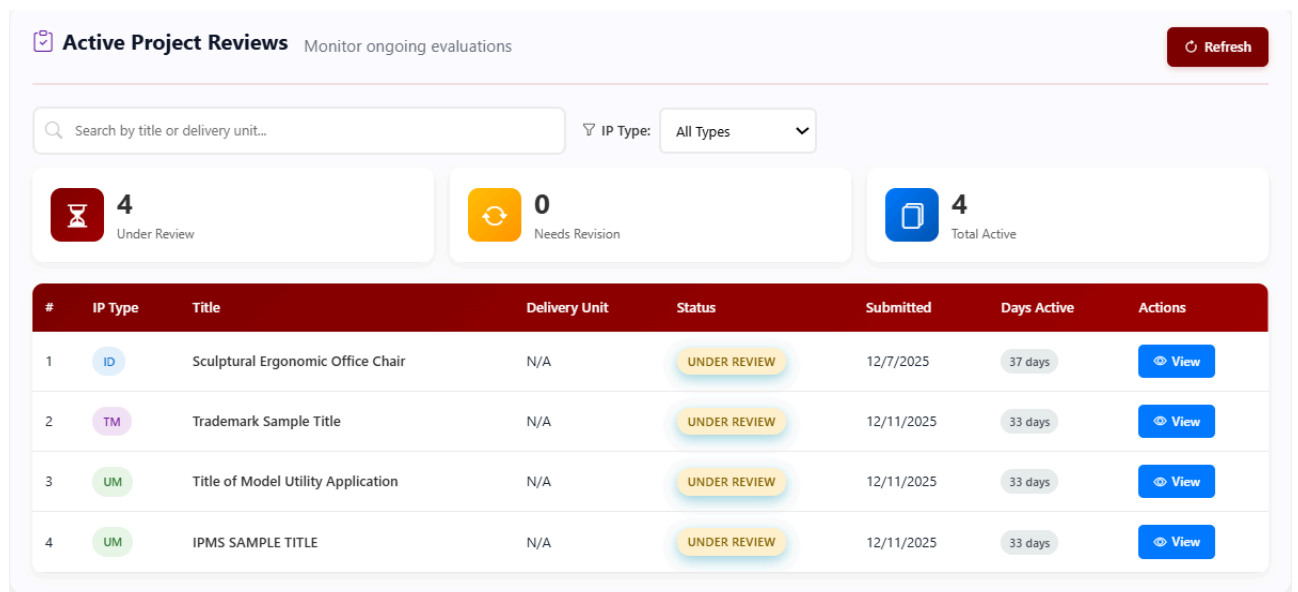


Figure 8. Project Submission Monitoring

Intellectual Property Reference Library. Read-only repository containing the official definitions and descriptions for each Intellectual Property category,

ensuring consistent standards as shown in Figure 9.

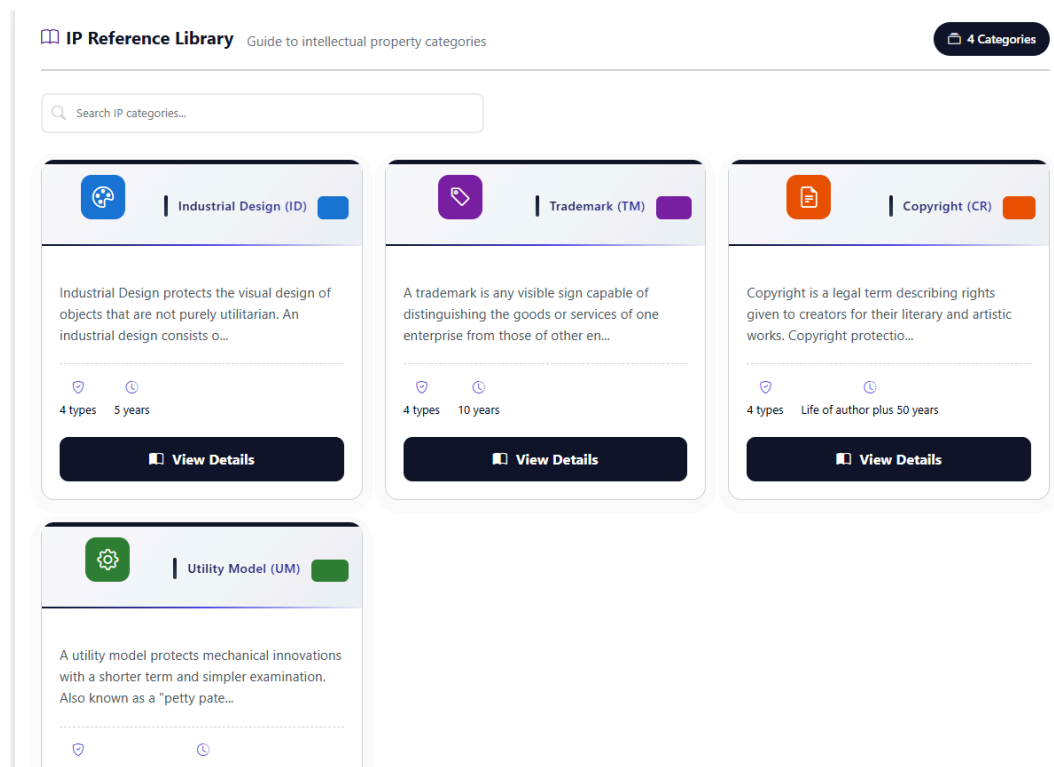


Figure 9. Intellectual Property Reference Library

Activity Logs. A feature that shows detailed records of every major action in the system- logins, uploads and status changes- for security and reporting purposes as shown in Figure 10.

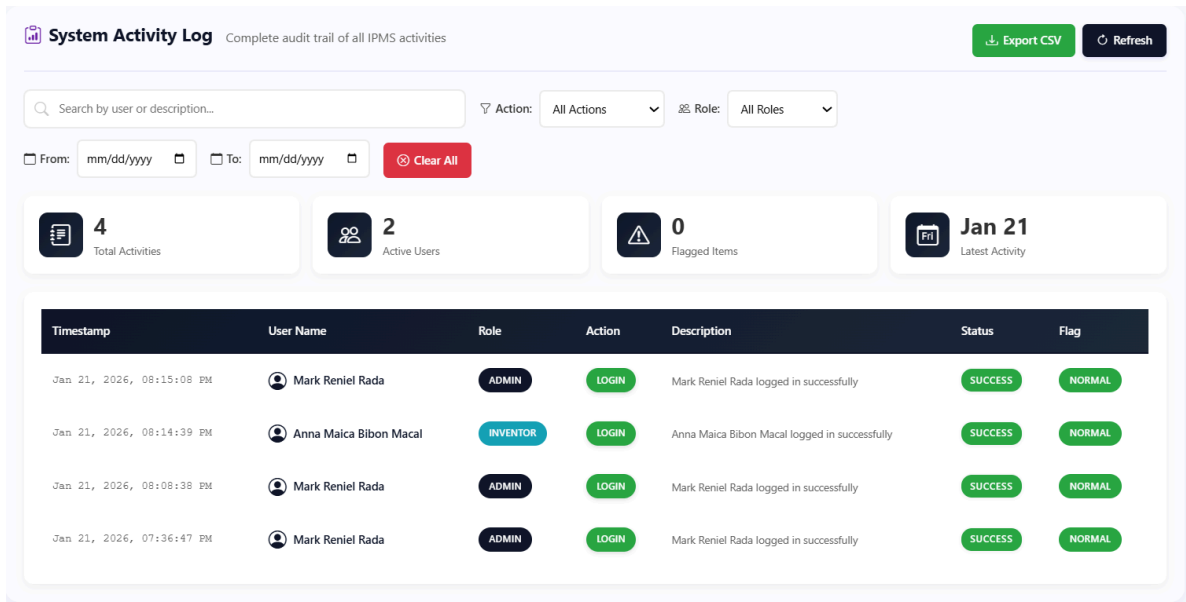


Figure 10. System Activity Logs

Open Review Queue. A list of all new submissions waiting for a technical check. Specialists can "receive" a project from this shared pool to begin their review as shown in Figure 11.

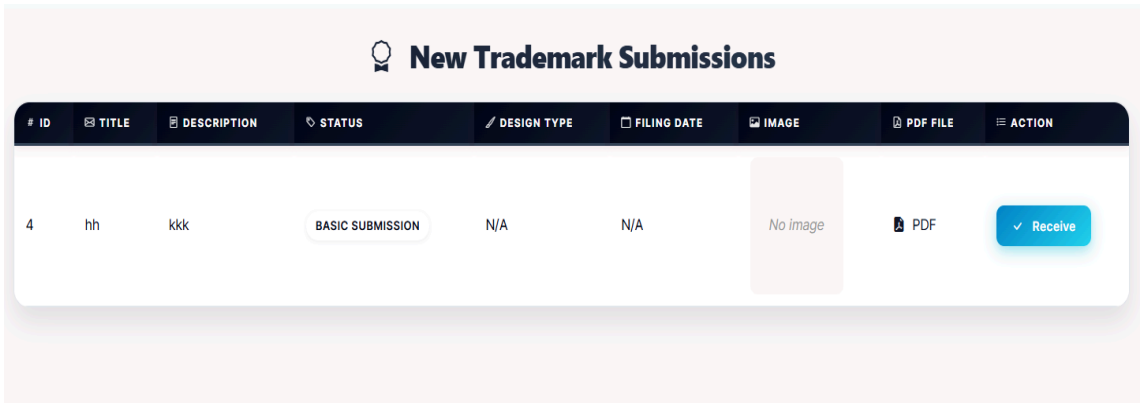


Figure 11. Open Review Queue

Feedback System. A communication Tool used to type detailed technical notes or requests or clarification back to the inventor if a submission is incomplete as shown in Figure 12.

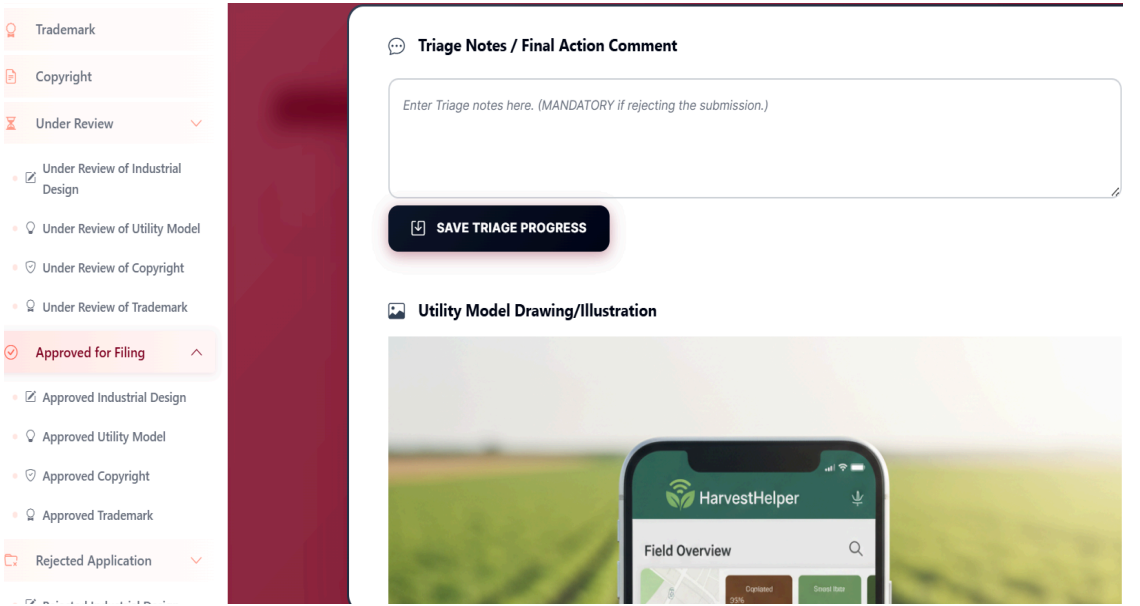


Figure 12. Feedback System

Decision Engine. A simple interface to mark a project as

Approved or Rejected, which instantly updates the Inventor's status tracker as shown in Figure 13.

Completeness Checklist (Triage)

Triage Status: **COMPLETE** (Ready for Final Review)

- ☒ Inventor contact details verified.
- ☒ Visuals are clear and all required views are present.
- ☒ Written description clearly identifies ornamental features.

Triage Notes / Final Action Comment

Enter Triage notes here. (MANDATORY if rejecting the submission.)

[Save Triage Progress](#)

[View File](#)

Official Form [View in New Tab](#) [Download](#)

IPOPHL Form 200

Intellectual Property Office of the Philippines
28 Upper McKinley Rd, Fort Bonifacio, Taguig City 1634 PH
☎ +63 (2) 7238-6300 | ✉ ask@ipophil.gov.ph

UTILITY MODEL REGISTRATION REQUEST

☐ Direct ☐ PCT ☐ Divisional ☐ w/ Claim of Priority

For IPOPHL use only	
Application No.	
Date Received	
Date Mailed	
IPSO / ITSO Code	

TITLE OF UTILITY MODEL
Utility Model Sample Form

DIVISIONAL INFORMATION (For Divisional Applications, if applicable)
Parent Application No. _____ Parent Application Filing Date (yyyy/mm/dd) _____

PCT INFORMATION (For National Phase Entry Applications, if applicable)
International Application Number _____ International Filing Date (yyyy/mm/dd) _____

[APPROVE FOR FILING](#) [REJECT SUBMISSION](#) [CLOSE](#)

Figure 13 Decision Engine

Inventor Portfolio Dashboard. A personalized workspace designed to empower inventors with real-time tracking of their specific intellectual property

applications. It features an interactive "Approved IP Submissions Trend" graph and color-coded summary cards that categorize approvals by type (e.g., Industrial Design, Trademark). The dashboard also includes a "My Submissions" feed, providing granular visibility into individual application statuses—such as "Under Review" or "New Submission"—allowing users to monitor progress and access application details effectively from a single interface as shown in Figure 14.

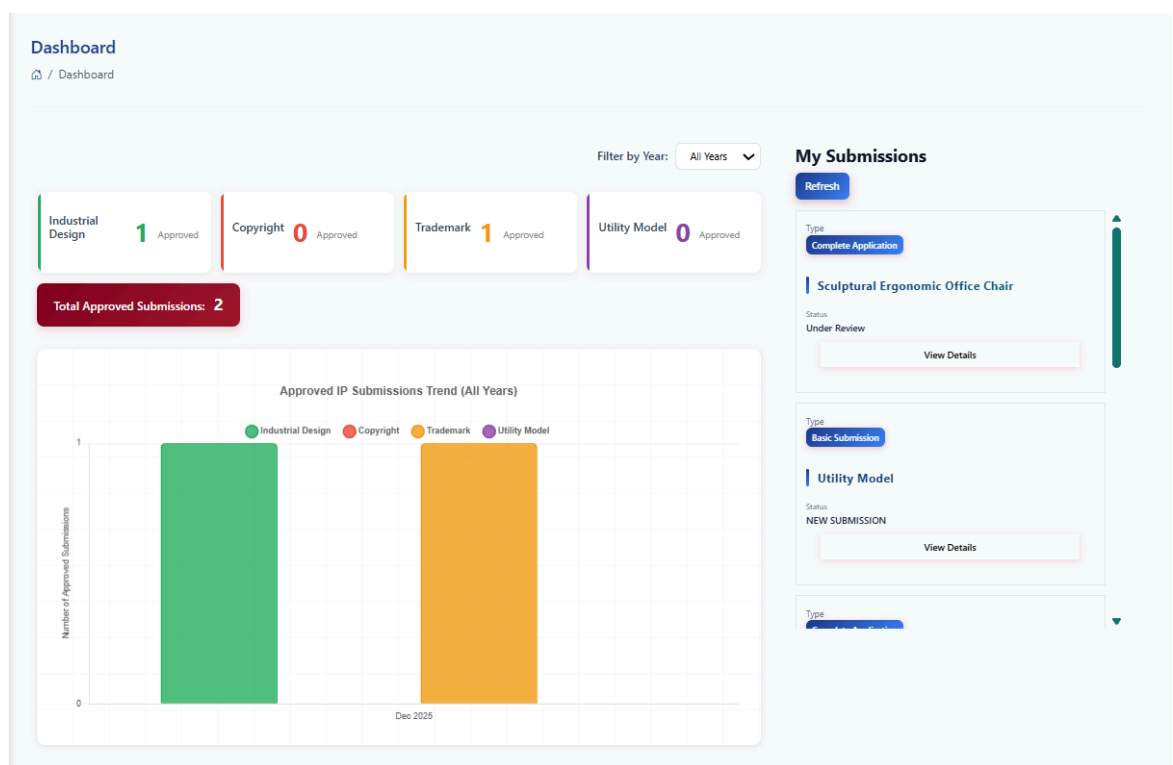
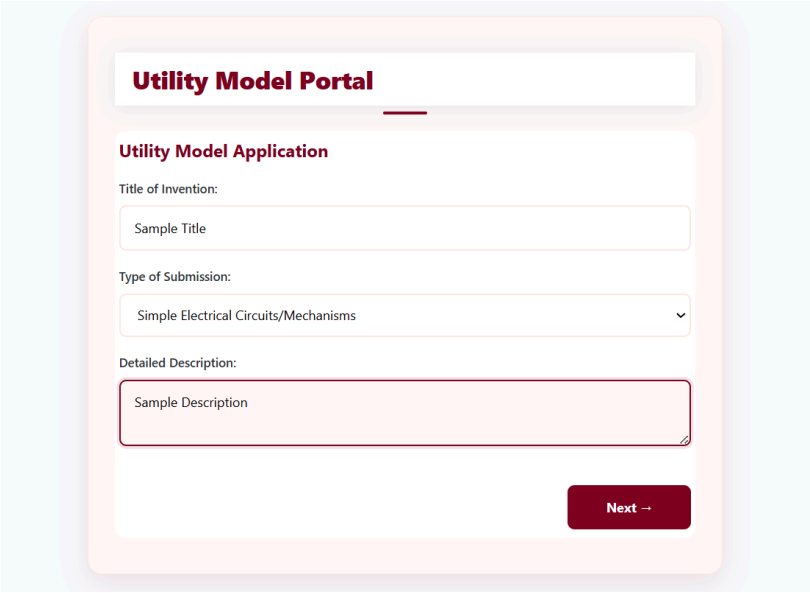


Figure 14 Inventors Portfolio Dashboard

Submission Portal. A guided digital platform where users enter detailed descriptions and select the type of their IP work, streamlining the application process through intuitive form fields and structured data entry. This interface replaces cumbersome paper-based

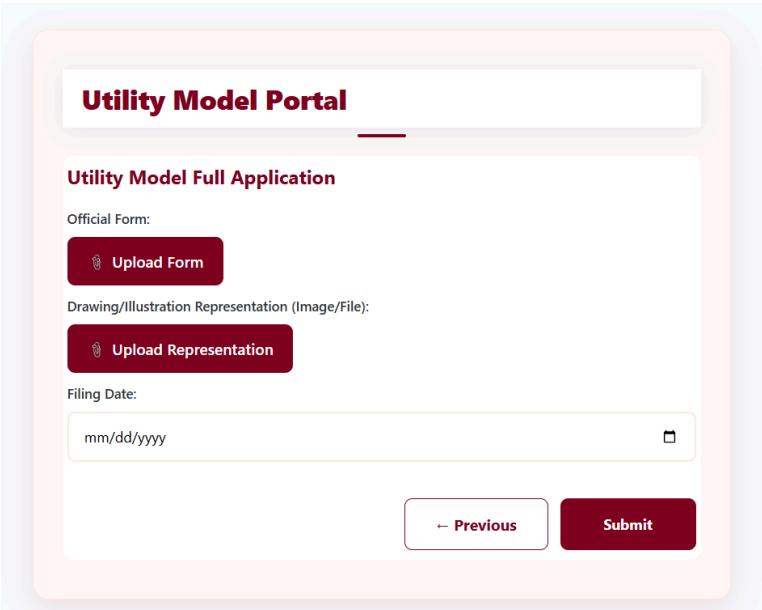
submissions with a user-friendly workflow that ensures completeness and consistency of submitted information, as shown in Figure 15.



The image shows a web form titled "Utility Model Portal" with a sub-header "Utility Model Application". It contains three input fields: "Title of Invention:" with a text box containing "Sample Title", "Type of Submission:" with a dropdown menu showing "Simple Electrical Circuits/Mechanisms", and "Detailed Description:" with a large text area containing "Sample Description". A "Next →" button is at the bottom right.

Figure 15: Submission Forms

File Uploader, A feature allowing the inventor to securely attach necessary supporting documents, image or source codes required for the specific IP category.



The image shows a web form titled "Utility Model Portal" with a sub-header "Utility Model Full Application". It contains three sections: "Official Form:" with an "Upload Form" button, "Drawing/Illustration Representation (Image/File):" with an "Upload Representation" button, and "Filing Date:" with a date input field showing "mm/dd/yyyy". At the bottom, there are two buttons: "← Previous" and "Submit".

Figure 16. File Uploader

Status Tracker. A feature showing the real-time progress of their submissions which can fall under these categories, Pending, Under Reviews, and Approved.

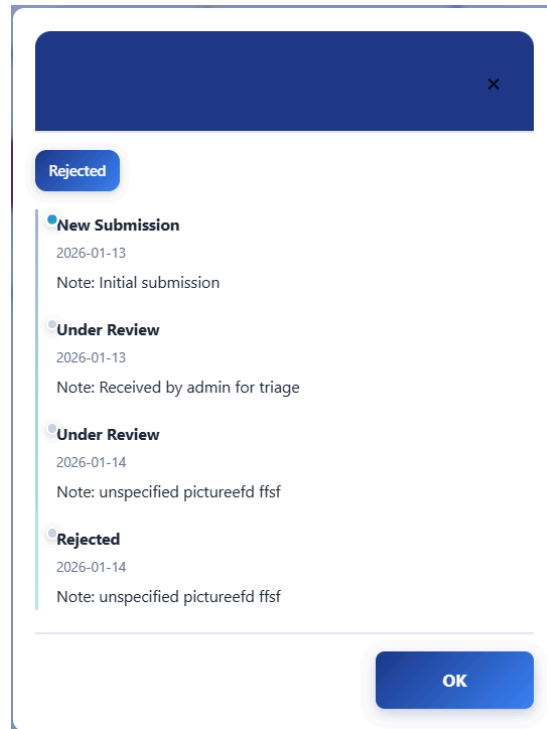


Figure 17. Status Tracker

Evaluation of ISO/IEC 25010 Quality Model

To answer the third research question regarding the level of effectiveness using the ISO/IEC 25010 Tool, the researchers surveyed 20 respondents consisting of 10 Students, 5 IPO-Phil submitters and 3 Office Staff of IPMO. Table 2 presents the results, showing that the IPMS performed excellently across all ISO 25010 subcategories. The scores for functionality suitability, efficiency, compatibility, usability and security were high, with the indicator rated at 4.75, confirming that the system is functional, secure and user-friendly.

Table 2

Weighted Mean per Subcategory of CNSC Students

ISO Category	25010 Main Subcategory	Weighted Mean
Functional Suitability	Functional Completeness	4.58
	Functional Accuracy	4.75
	Functional Management	4.58
Performance Efficiency	System Responsiveness	4.58
	System Scalability	4.75
	Transfer Efficiency	4.5
Compatibility	Device Compatibility	4.58
	File Compatibility	4.75
Usability	Interface Learnability	4.75
	User Guidance & Feedback	4.5
	UI Aesthetics	4.75
Security	Data Protection	4.75
	Control Management	4.66
	Information Security	4.66

As shown in Table 3, all ISO 25010 software quality characteristics for IPMS are rated by the Students as "Strongly Agree". Functional suitability, Performance Efficiency, Compatibility, Usability was 4.6 and Security at 4.69 proving that the system is reliable, secure, and easy to use. The overall weighted mean of 4.6 confirms that the system effectively meets the physician's

expectation in managing IP Documents. This also reflects the system's potential to further enhance the overall intellectual property document management process.

Table 3
Weighted Mean per Category of CNSC Students

ISO 25010 Category	Weighted Mean	Interpretation
Functional Suitability	4.6	Strongly Agree
Performance Efficiency	4.6	Strongly Agree
Compatibility	4.6	Strongly Agree
Usability	4.6	Strongly Agree
Security	4.69	Strongly Agree
Overall Weighted Mean	4.6	Strongly Agree

Table 4 shows the weighted mean of the scores for elements of the ISO 25010 software quality model. The categories are: functional suitability, functional efficiency, compatibility, usability, security and divided into subcategories each showing the performance based on user evaluation. Functional suitability, functional efficiency and security scored the highest of 5. Compatibility and usability scored 4.8. This indicates that users are highly satisfied with how well the system performs and how easy it is to use.

Table 4
Weighted Mean per Subcategory of IPO-Phil submitter

ISO	25010	Main	Weighted
Category		Subcategory	Mean
Functional Suitability		Functional Completeness	4.8
		Functional Accuracy	4.6
		Functional Management	5
Performance Efficiency		System Responsiveness	4.6
		System Scalability	5
		Transfer Efficiency	4.6
Compatibility		Device Compatibility	4.8
		File Compatibility	4.6
Usability		Interface Learnability	4.4
		User Guidance & Feedback	4.8
		UI Aesthetics	4.4
Security		Data Protection	5
		Control Management	5
		Information Security	4.8

With the total weighted mean of 4.7 and rated as "Strongly Agree", Table 5 shows that the IPO-Phil Submitter finds IPMS to be quite effective in handling the needed IP Document processing and it was relatively easy to navigate. Security scored the highest at 4.9, indicating that the users find the system to be secure and reliable while the lowest score is usability with the score of 4.5 while it is relatively high the user encountered minimal difficulty in navigating the system.

Table 5

Weighted Mean per Category of IPO-Phil submitter

ISO 25010 Category	Weighted Mean	Interpretation
Functional Suitability	4.8	Strongly Agree
Performance Efficiency	4.7	Strongly Agree
Compatibility	4.7	Strongly Agree
Usability	4.5	Strongly Agree
Security	4.9	Strongly Agree
Overall Weighted Mean	4.7	Strongly Agree

Table 6 shows that IPMS performs very well in all ISO 25010 subcategories. Compatibility and Usability obtained the highest ratings, particularly in Device/File compatibility and UI Aesthetics, which shows that the system is truly functional in supporting a wide range of Functional suitability and usability also scored high, demonstrating that the system is intuitive, easy to understand, easy to use, and easy to learn even for first-time users. The interface design prioritizes clarity and logical workflow, enabling users to navigate IPMS features with minimal guidance and training requirements. Respondents reported that they were able to perform tasks efficiently and accurately, with the system's organized layout and clear labeling reducing the learning curve significantly. This high usability rating

indicates that the IPMS successfully eliminates the confusion and frustration often associated with complex administrative systems, allowing users to focus on their primary objective of protecting intellectual property rather than struggling with the technology itself.

Table 6
Weighted Mean per Subcategory of Office Staff

ISO	25010	Main Subcategory	Weighted Mean
Functional Suitability		Functional Completeness	4.3
		Functional Accuracy	4.6
		Functional Management	4.6
Performance Efficiency		System Responsiveness	4.3
		System Scalability	4.3
		Transfer Efficiency	4.6
Compatibility		Device Compatibility	5.0
		File Compatibility	5.0
Usability		Interface Learnability	4.3
		User Guidance & Feedback	4.3
		UI Aesthetics	5.0
Security		Data Protection	4.3
		Control Management	4.6
		Information Security	4.3

Table 7 presents the results of the evaluation of the Office Staff on the proposed system, IPMS, that is based

on the ISO 25010 software quality model. IPMS obtained an overall weighted mean of 4.5 for "Strongly Agree". Compatibility scored the highest, 5. Functional Suitability and Usability both scored 4.5, Performance Efficiency and security 4.4.

Table 7

Weighted Mean per Category of Office Staff

ISO 25010 Category	Weighted Mean	Interpretation
Functional Suitability	4.5	Strongly Agree
Performance Efficiency	4.4	Strongly Agree
Compatibility	5.0	Strongly Agree
Usability	4.5	Strongly Agree
Security	4.4	Strongly Agree
Overall Weighted Mean	4.5	Strongly Agree

Table 8 compares ISO 25010 categories across groups of respondents-namely, functional suitability, functional Efficiency, Compatibility, Usability and security have been rated by the Office Staff of IPMO, IPO-Phil Submitters and Students. The result all shows very high ratings in all categories, with overall mean ranging from 4.4 to 5; all these are interpreted as "Strongly Agree". Every group of users states that the system to be highly functional, secure and efficiently user-friendly. Office Staff reports the scores a little lower than the other

demographics yet still pose a strong agreement. Overall, this indicates that the system is functional and secure yet user-friendly and capable of accurate support in a management of IP documents. These positive perceptions also mean that the system totally enhanced the general flow of works in the Intellectual Property Office.

Table 8

Overall Level of effectiveness per role

ISO 25010 Category	Student	IPO-Phil Submitters	Office Staff
Functional Suitability	4.6	4.8	4.5
Performance Efficiency	4.6	4.7	4.4
Compatibility	4.6	4.7	5.0
Usability	4.6	4.5	4.5
Security	4.69	4.9	4.4
Overall Weighted Mean	4.6	4.7	4.5
Interpretation	Strongly Agree	Strongly Agree	Strongly Agree

Figure 18 presents the ISO 25010 categories of Functional Suitability, Performance Efficiency, Compatibility, Usability, Security as evaluated by three groups of

respondents, namely Students, IPO-Phil Submitters, Office Staff. All categories yielded high ratings, with scores consistently falling within the range 4.4 and 5.0 for all respondents. Student and IPO-Phil Submitter Staff had the highest and most consistent ratings of over all weighted mean of 4.6 and 4.7 respectively, indicating strong agreement that the system is functional, secure and easy to use. Office Staff gave somewhat lower ratings of overall weighted mean of 4.5, a few points lower than the 2 groups, especially on Efficiency and Security but still a positive perception. The chart indicates a strong consensus among users representing various roles on the quality and effectiveness of the IPMS across key software quality characteristics. These consistent ratings suggest that the system is able to meet the different operational needs of every user of Intellectual Property Management System. Figure 18 shows the system's capabilities in advancing coordinated, efficient and reliable IP Management Office.

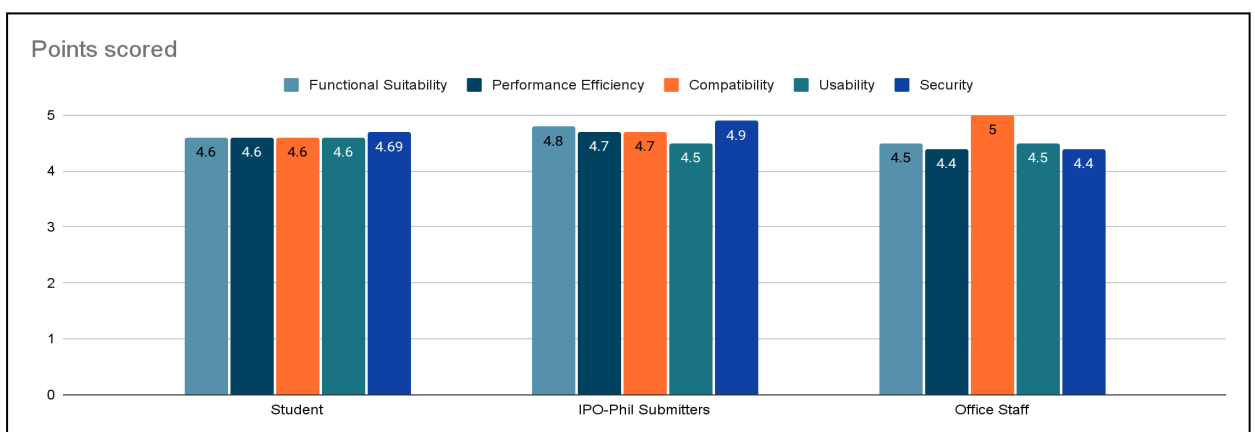


Figure 18. Overall Level of Effectiveness per Role

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the conclusions drawn from the findings, followed by recommendations for implementation and future development.

1. Current System Assessment (Research Objective 1)

The investigation of the current system at the Intellectual Property Management Office (IPMO) revealed that the office operates predominantly through a manual, paper-based system for processing intellectual property applications and managing records. This traditional approach, while providing legal documentation and familiarity for existing staff, presents significant operational challenges. The manual system is characterized by physical document storage in filing cabinets, limited digital tool integration, and time-consuming manual review processes. These limitations result in inefficiencies including delayed processing times, increased risk of data loss or damage, difficulty in tracking application status, and constrained scalability as submission volumes increase. The findings align with existing literature demonstrating that manual IP management systems experience data inconsistencies, service delays, and communication breakdowns between inventors and reviewers. The absence of centralized

digital repositories creates vulnerabilities in data security and complicates compliance auditing, exposing the institution to potential legal and administrative risks.

2. System Features and Innovations (Research Objective 2)

The developed Intellectual Property Management System (IPMS) successfully integrates a comprehensive suite of core functionalities and technological innovations designed to address the limitations of the current manual system. The IPMS features a centralized login interface with role-based access control for Administrators, Specialists, and Inventors, ensuring secure and appropriate system access. Key innovations include an Admin Dashboard providing real-time system metrics and performance monitoring, a User Management module for centralized account control and authentication, and a Project Submission Monitoring feature ensuring no submission is overlooked. The system incorporates an Intellectual Property Reference Library maintaining consistent standards across all IP categories, along with detailed Activity Logs for security and accountability.

For Specialists, the IPMS provides an Open Review Queue streamlining the evaluation process, an integrated Feedback System enabling direct communication with inventors, and a Decision Engine facilitating efficient

approval or rejection workflows. Inventors benefit from a personalized Portfolio Dashboard offering real-time tracking of application progress, a guided Submission Portal replacing cumbersome paper-based processes, a secure File Uploader supporting various document types, and a Status Tracker providing transparent visibility into application stages. These features collectively transform the IP management workflow from a fragmented, manual process into an integrated, digital ecosystem that enhances efficiency, transparency, and user experience while maintaining data integrity and security.

3. System Effectiveness Evaluation (Research Objective 3)

The evaluation of the IPMS using the ISO/IEC 25010 quality model demonstrated exceptionally high levels of effectiveness across all assessed characteristics. The system was evaluated by 20 respondents comprising 10 students, 5 IPO-Phil submitters, and 5 office staff from IPMO. Results consistently showed weighted means ranging from 4.4 to 5.0 across all categories, with all groups rating the system as "Strongly Agree" for overall effectiveness.

Functional Suitability received high ratings across all user groups (Students: 4.6, IPO-Phil Submitters: 4.8, Office Staff: 4.5), confirming that the system effectively meets its intended purpose with complete, accurate, and appropriate functionality. Performance

Efficiency scores (Students: 4.6, IPO-Phil Submitters: 4.7, Office Staff: 4.4) demonstrate that the system utilizes resources effectively while maintaining responsiveness and scalability. Compatibility ratings were particularly strong (Students: 4.6, IPO-Phil Submitters: 4.7, Office Staff: 5.0), with Office Staff giving the highest score, indicating excellent device and file format compatibility.

Usability evaluations (Students: 4.6, IPO-Phil Submitters: 4.5, Office Staff: 4.5) confirm that the interface is learnable, provides adequate guidance and feedback, and maintains aesthetic appeal. The UI design successfully prioritizes clarity and logical workflow, enabling users to navigate system features with minimal training requirements. Security assessments yielded consistently high scores (Students: 4.69, IPO-Phil Submitters: 4.9, Office Staff: 4.4), with IPO-Phil Submitters rating it highest, demonstrating strong confidence in the system's data protection capabilities, access control management, and information security measures.

The overall weighted means across user groups (Students: 4.6, IPO-Phil Submitters: 4.7, Office Staff: 4.5) all fall within the "Strongly Agree" range, validating that the IPMS successfully addresses the operational challenges identified in the current manual

system while meeting the diverse needs of all stakeholder groups. These results confirm that the system is functional, secure, user-friendly, and capable of providing accurate support in the management of intellectual property documents.

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APPENDICES

Appendix A

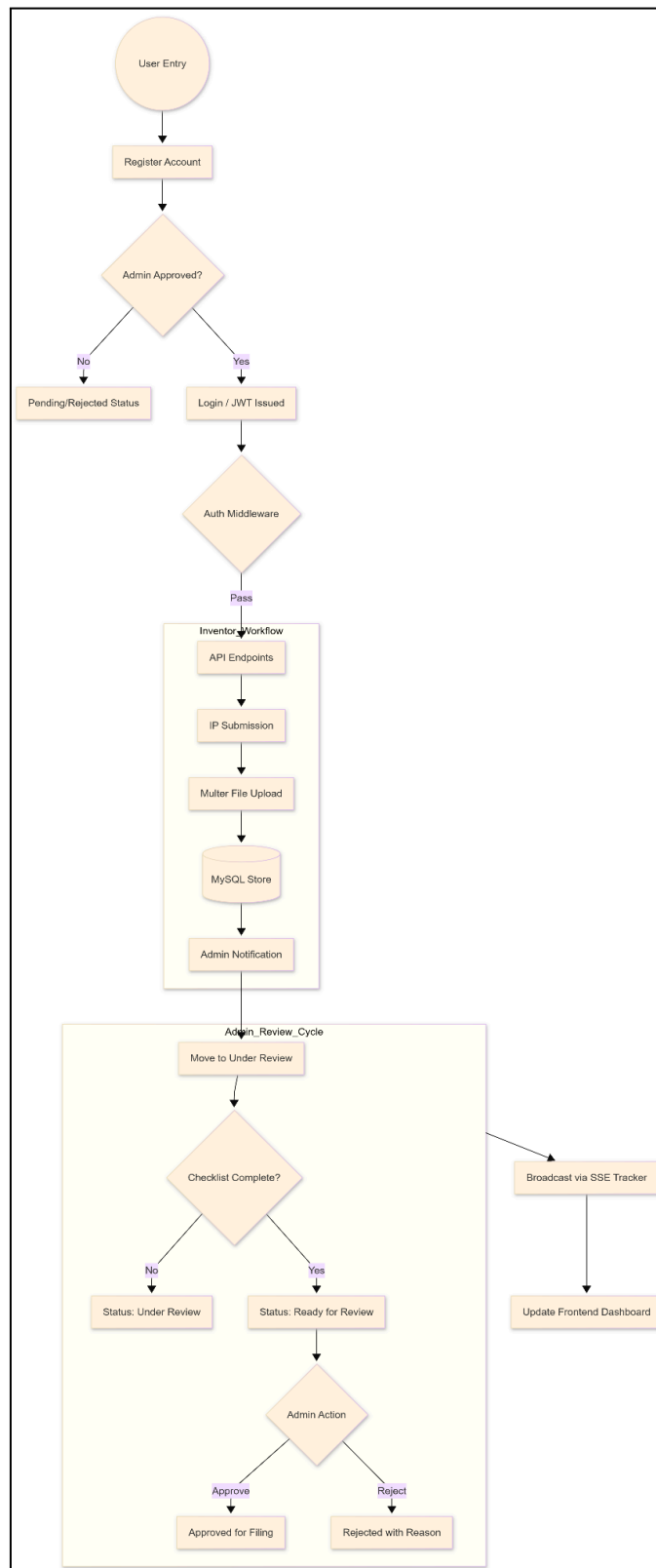
Interview Guide

No.	Questions
1	Currently, how do applicants submit their filings?
2	When a new application comes in, how do you verify that all required documents (drawings, descriptions, fee receipts) are included before it reaches an examiner?
3	Is it a manual verification process, or is it automatically updated in your records?
4	Do applications for Copyright, Trademarks, and Utility Models, Industrial Design follow the same intake path, or are they handled by entirely different systems/teams from the start?
5	On average, how much time does an examiner spend just searching for or "checking out" the physical or digital files needed for a single review?
6	If an applicant sends a corrected version of a design or a claim, how do you ensure the examiner is looking at the most recent version and not an outdated draft?
7	How are "Office Actions" (notices of deficiencies or rejections) sent to the applicant? (Email, automated system, or registered mail?)
8	Is there an automated alert system for pending deadlines?
9	When an applicant resubmits a corrected document, is it automatically "attached" to the original case file, or does a clerk have to manually pair them up again?
10	Can applicants check the status of their application online 24/7, or do they have to call or visit the office for updates?
11.	"When multiple people (e.g., an examiner and a legal officer) need to discuss a specific application, how is that conversation recorded?"
12.	"What percentage of applications are delayed or sent back solely due to simple clerical errors (like missing signatures or incorrectly filled fields)? Would a system

	that 'stops' the user from submitting until everything is correct save significant staff time?"
13.	"How do supervisors currently monitor the workload of the examiners? Is there a way to see at a glance how many files are 'Overdue' or 'In Progress' across the entire office, or is that information compiled manually?"
14.	How do you trace the history of that specific document? Is there a digital audit trail that records every 'touch' on a file?"
15.	If you could automate or digitize just <i>one</i> step in the current process to make your life 50% easier, which one would it be?

Appendix B

User Work Flow of IPMS



Appendix C

ERD of IPMS



Appendix D

Core Module

Appendix E

Testing Phase of the System

A. TESTING FOR INVENTORS

Test Case ID	Test Case Name	Description	Test Data / Input	Expected Result	Actual Result	Remarks
1	Login to System	Verify Applicant can log in with valid credentials	Valid email & password	Redirected to Applicant Dashboard	Redirected to Applicant Dashboard	PASS
2	Invalid Login Attempt	Check behavior when incorrect password is entered	Invalid password	"Invalid credentials" message appears	"Invalid credentials" message appears	PASS
3	Submit New Filing	Confirm Applicant can submit a new IP application (e.g., Trademark)	Complete form data & upload required PDF	"Application Submitted Successfully" modal appears	"Application Submitted Successfully" modal appears	PASS
4	View Application Status	Ensure Applicant can track the status of their submission	Click "My Applications" tab	Displays list with status "Pending Review"	Displays list with status "Pending Review"	PASS

B. TESTING FOR CONSULTANT

Test Case ID	Test Case Name	Description	Test Data / Input	Expected Result	Actual Result	Remarks
1	Login to System	Verify Examiner can log in with valid credentials	Valid staff ID & password	Redirected to Examiner Dashboard	Redirected to Examiner Dashboard	PASS
2	View Pending Queue	Verify Examiner can see list of new submissions	Navigate to "Incoming Applications"	List of new applications is displayed	List of new applications is displayed	PASS

3	Review Document	Check if Examiner can open and view uploaded documents	Click on specific Application ID	Document viewer opens with split-screen view	Document viewer opens with split-screen view	PASS
4	Send Office Action	Test if Examiner can send a correction request/feedback	Input comments & click "Request Correction"	Status changes to "For Resubmission" & alert sent	Status changes to "For Resubmission" & alert sent	PASS

C. TESTING FOR ADMINISTRATOR

Test Case ID	Test Case Name	Description	Test Data / Input	Expected Result	Actual Result	Remarks
1	Login to System	Verify Admin can log in with valid credentials	Valid Admin ID & password	Redirected to Admin Dashboard	Redirected to Admin Dashboard	PASS
2	User Management	Confirm Admin can add a new Examiner account	Input new staff details & set role	"User Created Successfully" message	"User Created Successfully" message	PASS
3	Generate Reports	Verify Admin can export monthly application statistics	Select date range & click "Export PDF"	PDF Report downloaded automatically	PDF Report downloaded automatically	PASS
4	Backup Data	Check if manual backup function works	Click "Backup Database" button	"Backup Completed" notification appears	"Backup Completed" notification appears	PASS

Appendix F

Collected Bugs & Issues

Appendix G

System Deployment Preparation

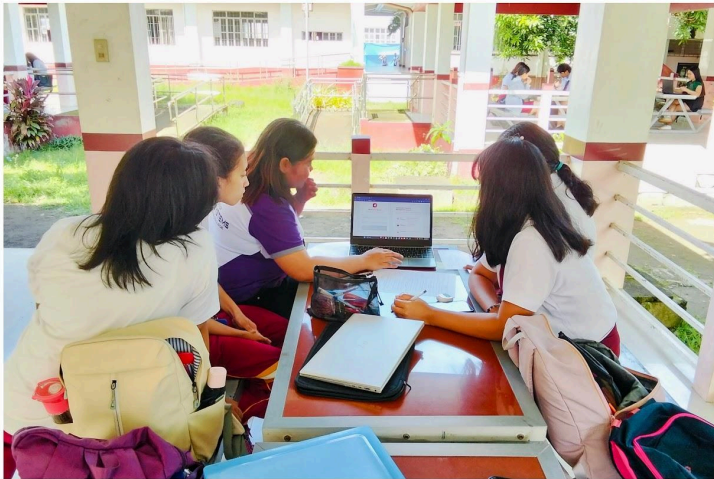
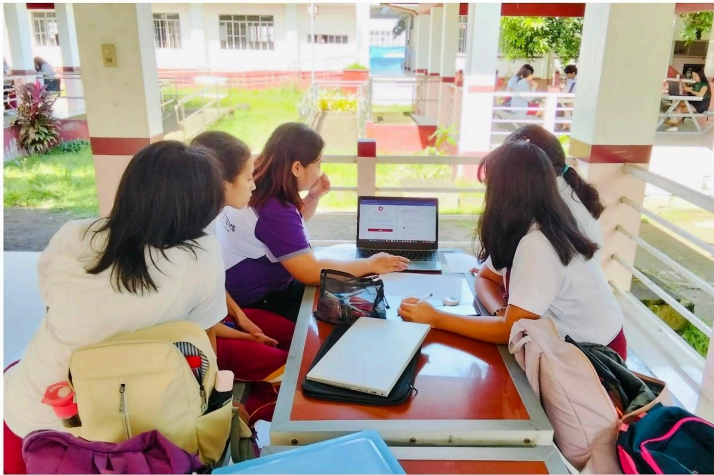
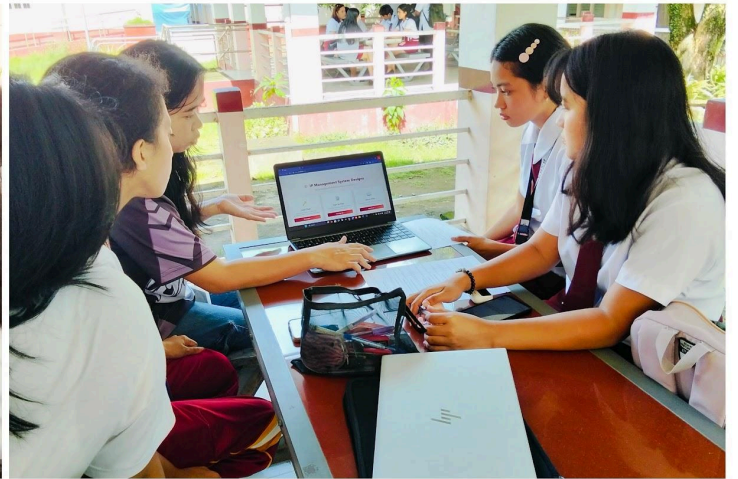
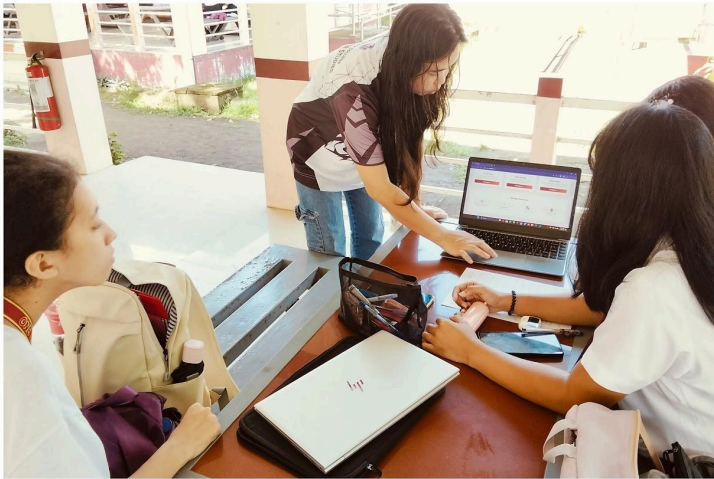
Appendix H

Sample Questionnaire per role

Appendix I

System Evaluation





Appendexis J

System Development Work Session